Naloxone

Mechanism of action

- Naloxone is an opiate receptor antagonist (blocker). By blocking opiate receptors, naloxone reverses the effects of opiates, particularly respiratory depression and sedation.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Opiate poisoning is suspected and the patient has a significantly impaired level of consciousness or significantly impaired breathing.
- ✓ Excess adverse effects from administration of opiates.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Chronic opiate use. If the patient is taking an opiate chronically, there is a risk of adverse physiological effects associated with rapid opiate withdrawal.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated in pregnancy, but naloxone should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 0.1-0.4 mg IV every five minutes as required for an adult.
- 0.8 mg IM as required for an adult. This may be repeated every 10 minutes.
- See the paediatric drug dose tables for a child.

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Administration

- The preferred route for administration is IV.
- IV: dilute 0.4 mg to a total of 4 ml. This final solution contains 0.1 mg/ml.
- Administer the minimum dose required to produce improvement. Rapid reversal of opiates may be associated with seizures, hypertension, pulmonary oedema or severe agitation, particularly if the patient takes opiates regularly.
- IM: administer undiluted. The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

Common adverse effects

- Sweating.
- Tachycardia.
- Hypertension.

Usual onset of effect

- IV: 1-2 minutes.
- IM: 5-10 minutes.

Usual duration of effect

- 30-60 minutes.
- The duration of action of naloxone may be shorter than the duration of action of the opiate that has been administered/taken and naloxone may need to be repeated.

Usual preparation

- Ampoule containing 0.4 mg in 1 ml.

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Pharmacokinetics

- Naloxone is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- None.

Additional information

- There is no role for naloxone in the treatment of cardiac arrest associated with opiate poisoning. In this setting cardiac arrest is secondary to respiratory arrest and once cardiac arrest has occurred naloxone has no useful effect. The best treatment is CPR that includes a focus on ventilation. If ROSC occurs, naloxone should still not be administered because it may be associated with seizures, hypertension, pulmonary oedema or severe agitation.
- There is no evidence to support the commonly held view that adequate oxygenation prior to naloxone administration reduces the severity of agitation following naloxone administration. However, treatment of severe hypoxia takes precedence over the administration of naloxone.
- Acute opiate withdrawal following naloxone administration can lead to patient agitation and/or aggression, particularly when there has also been a period of hypoxia. Ensure a planned team approach to maintain safety of both the patient and personnel.